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Scalable Ticket Booking System Design

1. Functional Requirements (FR's)

- **Search & Discovery:** Users must search for events by title, city, or genre with fuzzy matching.
- **Seat Selection:** Real-time visualization of seat maps with current availability status.
- **Temporary Seat Locking:** Atomic locking of selected seats for 5 minutes to prevent race conditions.
- **Booking & Payment:** Secure transaction processing with automated ticket generation upon success.
- **Async Notifications:** Delivery of booking confirmation via Email/SMS post-transaction.

2. Non-Functional Requirements (NFR's)

- **High Availability:** The browsing/searching flow must be 99.99% available.
- **Strong Consistency:** Mandatory for the booking and seat-locking flow to prevent double bookings.
- **Low Latency:** Search results should return in <100ms; seat locking should occur in <200ms.
- **Scalability:** Ability to handle massive traffic spikes (100k+ TPS) during popular event releases.
- **Fault Tolerance:** Use of circuit breakers for third-party payment gateways to prevent cascading failures.

3. API Design

- `GET /v1/events?city={id}&q={query}`: Returns filtered events from ElasticSearch.
- `GET /v1/shows/{show_id}/seats`: Fetches real-time seat map from Read Replicas/-Cache.
- `POST /v1/seats/lock`: (Payload: `seat_ids[]`) Initiates Redis `SETNX` with a 300s TTL.
- `POST /v1/bookings`: Creates a pending order and initiates the payment redirect.
- `POST /v1/payments/webhook`: Asynchronous callback endpoint for payment gateway confirmation.

4. Core Entities

- **User:** Profile, authentication credentials, and preferences.
- **Event/Show:** Movie/Concert metadata, venue mapping, and timing.

- **Seat:** Physical coordinate, category (VIP/Gold), and price.
- **Booking/Order:** Transactional record linking User, Show, and Seats.
- **SeatLock:** Temporary distributed lock entity (Redis-backed).

5. Database Schema Design

User Table: (user_id, name, email, phone)
Show Table: (show_id, event_id, venue_id, start_time, base_price)
Seat Table: (seat_id, show_id, seat_num, status [Available, Locked, Booked])
Booking Table: (booking_id, user_id, show_id, total_amt, status [Pending, Paid, Failed])
Outbox Table: (event_id, payload, status) – *Used for Transactional Outbox Pattern to ensure Kafka sync.*

6. High Level Design (HLD)

The architecture follows a microservices pattern optimized for high-concurrency:

- **Global Edge:** CDN handles static assets; L7 Load Balancer manages traffic across API Gateways.
- **Concurrency Layer:** **Redis SETNX** acts as a distributed lock. If payment fails or the 5-min TTL expires, the lock is released automatically.
- **Search Strategy:** A **CDC (Change Data Capture)** pipeline via Kafka Connect streams SQL updates to **ElasticSearch** for sub-second searching.
- **Reliability:** The **Transactional Outbox Pattern** ensures that the Booking DB update and the Notification Service (Kafka) are always in sync.
- **Data Strategy:** Write-intensive booking operations hit the **Primary SQL DB**, while heavy seat-map reads are distributed across **Read Replicas**.

Figure 1: High Level Design of the Scalable Ticket Booking System